Manasvi Gampa

me23btech11035@iith.ac.in | manasvi.gampa@gmail.com | +91 8143955378 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.36
FIITJEE Junior College	Class XII (TSBIE)	2023	97.60%
IRIS World School	Class X (ICSE)	2021	93.00%

SCHOLASTIC ACHIEVEMENTS

- Solved 750+ problems across LeetCode (Rated 1680, 400+), GeeksforGeeks (170+), Codeforces and CodeChef.
- ICSE Skill Olympiad: Ranked 17th nationally
- TEDEd Student Talks: Presented an original talk under TEDEd's global youth initiative
- SOF International Mathematics Olympiad: Gold Medalist (School Level)
- ISRO Cyberspace Competitions 2020: Ranked among Top 500 nationwide (Essay Category)

WORK EXPERIENCE

Chubb | *Technology Intern*

- Developed the Workflow Configuration module for DocAI Intake V2, enabling dynamic creation and management of document-processing workflows.
- Designed and implemented a layered Spring Boot backend with Entity, Repository, DTO, Service, and Controller components, using PostgreSQL/JSONB for configurable workflow-step persistence.
- Built and integrated 8 REST APIs with the Admin UI, enabling seamless workflow creation, retrieval, step configuration, toggling, and plugin assignment.
- Contributed to V2's dynamic configuration model, reducing new LOB/region onboarding from 4-8 weeks in V1 to ~1 sprint by empowering domain teams to configure workflows independently.

PROJECTS

RAGBench: Document Intelligence Platform | *Python, LangChain, ChromaDB, HuggingFace, Gradio*

- Built a modular RAG pipeline supporting document directories with recursive ingestion and configurable chunking.
- Implemented token-aware preprocessing, semantic embeddings, ChromaDB retrieval, and conversational LLM generation, with configurable retrieval and context limits.
- Developed a Gradio interface with session-isolated vector stores, processing logs, chat memory, and page-level source attribution for retrieved documents.

PostgreSQL Query Performance & Optimizer Analyzer | *Python, PostgreSQL, pg_hint_plan*

- Benchmarked 5 indexing strategies on a 60M-row TPC-H dataset, reducing query runtime by 2.5× via optimal clustered indexing at moderate selectivities.
- Automated plan-equivalence and switch detection across 3 parameterized queries, identifying 14 optimizer plan switches.
- Exposed suboptimal PostgreSQL query routing (early/delayed plan switches) by enforcing alternative execution paths using pg_hint_plan.

Geospatial Clustering with Custom K-Means | *Python, NumPy, Pandas*

- Analyzed Telangana pincode data and implemented K-Means clustering from scratch, identifying 20 geographic clusters.
- Applied spatial clusters to logistics, resource allocation, and urban planning.

Sentiment Analysis of Handwritten Text Using OCR | *Python, CNN, OCR*

- Built a handwritten sentiment pipeline using CNN-based OCR and Naive Bayes, without pre-trained models.
- Enabled sentiment extraction from offline handwriting for assistive technology, screening, and feedback analysis.

SKILLS

Programming Languages: Java, Python, C, C++

Data Science & ML Libraries: NumPy, Pandas, Matplotlib, SciPy, scikit-learn, TensorFlow

Databases: SQL, MySQL, PostgreSQL

Tools & Platforms: Jupyter Notebook, LaTeX, Visual Studio Code, IntelliJ IDEA, Google Colab, Microsoft Office

Soft Skills: Strong analytical skills, effective communication, compelling presentation abilities

RELEVANT COURSES

Computer Science:

- *College:* Introduction to Programming, Introduction to Modern AI, Deep Learning, Database Management Systems, Computer Architecture
- *Independent Learning:* Data Structures & Algorithms, Object-Oriented Programming (OOP)

Mathematics: Elementary Linear Algebra, Probability, Statistics

POSITIONS OF RESPONSIBILITY

- TEDxIITHyderabad: Content & Curation Coordinator
- International Relations Cell: Inbound Buddy
- Aero (Aeronautics & Drone Design Club): Core Member

EXTRACURRICULAR

- Member of the NSO Athletics team; specialized sprinter with podium finishes at school and college-level events.
- Enjoy creating detailed pencil sketches and intricate Zentangle art, as a medium of self-expression and mindfulness.
- Passionate about swimming, with accolades earned at school-level tournaments and college fests.
- Avid follower of global cinema and TV, with a strong appreciation for English, Korean, Chinese, and Japanese narratives.